

Heart Disease Prediction Using Decision Trees: A CRISP-DM Based Approach with Clinical Interpretability

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Abstract—This paper presents a comprehensive approach to heart disease prediction using decision tree classification, specifically implementing the CART (Classification and Regression Trees) algorithm. Using a dataset of 918 patient records with 11 clinical features, we achieved 88% prediction accuracy while maintaining medical interpretability. The study employs the CRISP-DM framework and compares entropy and gini-based splitting criteria, finding that gini-based splitting with optimized tree depth of 4 provides the best balance between accuracy and interpretability. Our results identify ST-Slope (0.643), Chest-PainType (0.091), and Cholesterol (0.083) as the most significant predictors, aligning with clinical knowledge. The model's performance and transparent decision-making process make it particularly suitable for clinical decision support applications.

I. INTRODUCTION

Heart disease remains a leading cause of mortality worldwide, making early and accurate prediction crucial for effective intervention. While various machine learning approaches exist for heart disease prediction, the challenge lies in balancing predictive accuracy with model interpretability, particularly in clinical settings where understanding the decision-making process is crucial.

A. Research Questions

This study addresses three key questions:

- How can CART algorithms be optimized for heart disease prediction while maintaining clinical interpretability?
- What is the relative importance of different clinical features in prediction accuracy?
- How do different tree optimization strategies affect model performance and interpretability?

II. LITERATURE REVIEW

A. Machine Learning in Cardiovascular Prediction

Recent studies have shown increasing adoption of machine learning in cardiovascular disease prediction. Wu et al. [1] conducted a comprehensive review highlighting decision trees' particular advantage in medical applications due to their interpretability. Their findings support our choice of CART algorithm and focus on maintaining transparency in the decision process.

B. Feature Selection and Clinical Relevance

Ahmad et al. [2] demonstrated the critical importance of feature selection in cardiovascular prediction models. Their work identified several key clinical indicators, particularly noting the significance of ST segment analysis, which aligns with our findings regarding ST-Slope importance.

C. Model Interpretability in Healthcare

Singh and Singh [3] compared various algorithms for heart disease prediction, emphasizing the trade-off between model complexity and interpretability. Their research influenced our approach to hyperparameter optimization and tree depth limitation.

III. METHODOLOGY

A. CRISP-DM Framework Implementation

Our methodology follows the CRISP-DM framework:

- **Business Understanding:** Identified need for interpretable heart disease prediction models
- **Data Understanding:** Analyzed 918 patient records with 11 clinical features
- **Data Preparation:** Implemented feature encoding and scaling
- **Modeling:** Developed and optimized decision tree classifier
- **Evaluation:** Assessed performance through multiple metrics
- **Deployment:** Focused on clinical applicability

B. Dataset Description

The study utilized the Heart Disease Prediction Dataset comprising:

- 918 patient records
- 11 clinical features including demographic and medical measurements
- Binary target variable indicating heart disease presence
- Complete data with no missing values

C. Data Preprocessing

Our preprocessing pipeline included:

- Categorical variable encoding using Label Encoder
- Numerical feature standardization
- Feature quality verification
- Train-test split with stratification

D. Model Development

The decision tree implementation included:

- Grid search optimization for hyperparameters
- Comparison of entropy and gini criteria
- Cross-validation for model validation
- Tree depth limitation for interpretability

IV. RESULTS AND DISCUSSION

A. Model Performance

Our optimized model achieved:

- Overall accuracy: 88%
- Training accuracy: 87%
- Test accuracy: 88%
- Balanced performance across classes

B. Feature Importance Analysis

Key predictive features identified:

- ST-Slope (0.643 importance score)
- ChestPainType (0.091 importance score)
- Cholesterol (0.083 importance score)

C. Optimal Model Parameters

Best parameters found through grid search:

- Criterion: Gini
- Maximum depth: 4
- Minimum samples leaf: 4
- Minimum samples split: 2

D. Cross-validation Results

Five-fold cross-validation showed:

- Mean accuracy: 0.808
- Standard deviation: 0.109
- Consistent performance across folds

E. Clinical Relevance

The model demonstrates strong clinical relevance through:

- Interpretable decision paths
- Feature importance alignment with medical knowledge
- Balance between accuracy and explainability

V. CONCLUSION AND FUTURE WORK

This study successfully demonstrates the effectiveness of optimized CART algorithms in heart disease prediction, achieving 88% accuracy while maintaining clinical interpretability. The identified feature importance hierarchy aligns with medical knowledge, making the model suitable for clinical decision support.

Future work could explore:

- Integration of additional clinical features
- Ensemble methods while maintaining interpretability
- Real-world clinical validation studies
- Dynamic feature importance analysis

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